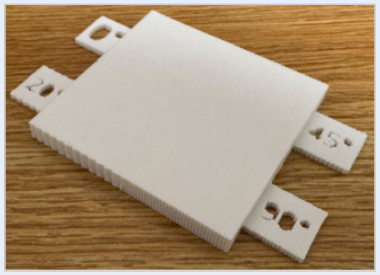
Heewon Park

Biomedical Engineering | Optics, Biosensing & Medical Device R&D

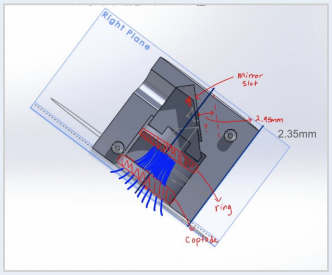
I'm a Master of Engineering student in Biomedical Engineering at Boston University. My work spans optical imaging phantoms, optical probe design, and software tools for biological data. Each project below is a short summary of the problem, what I set out to do, what I did, and what happened.



Anisotropic Phantoms for Clinical Optical Imaging

A 3D-printed skin-mimicking phantom measured with SFDI.

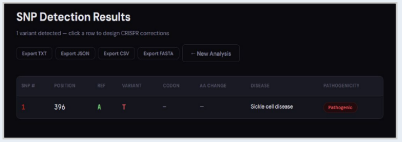
Page 2



Optical Probe (Optode) Casing

A SolidWorks casing for a small fiber bundle that measures blood flow through hair.

Page 3



Cleave SNP: SNP Detection and CRISPR Guide Design

A web platform that detects a sickle cell SNP and ranks guide RNAs.

Page 4



Hangle: Hangman Meets Wordle

A Hangman and Wordle web game built by a team of six.

Page 5

ALSO INCLUDED

Page 6: an Arduino room occupancy monitor and a human-subject caffeine study.

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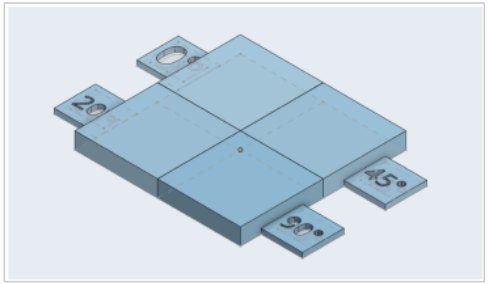
heewon@bu.edu | linkedin.com/in/heewon-park-

Fabrication of Reproducible Anisotropic Phantoms

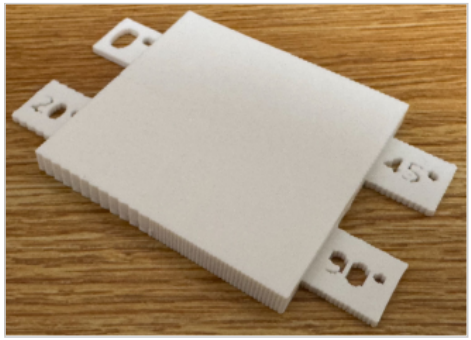
Boston University · Team of 4 · Sep 2025 – May 2026 | Optical imaging · 3D printing · SFDI

Scleroderma hardens skin through fibrosis, replacing normal basket-weave collagen with aligned collagen bundles, and we hypothesized that this makes skin optically anisotropic. Because no phantoms or test methods existed to check anisotropy, optical tools such as SFDI were hard to test and calibrate. Our team of four set out to build a phantom with skin-like optical properties and tunable anisotropy angles that SFDI could measure, judged by whether its measurements matched known dermis values, showed detectable anisotropy, and were reproducible across samples.

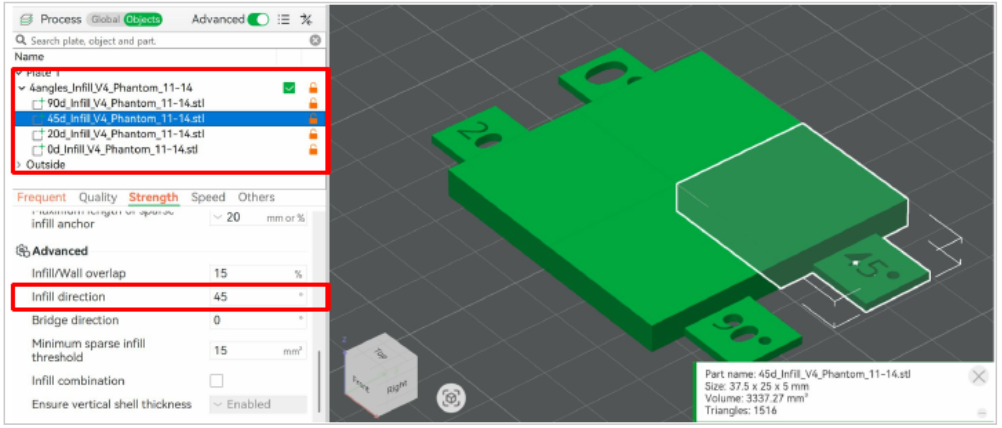
We reviewed the literature, designed the phantom in CAD, set infill directions in Bambu Studio, and 3D-printed about 20 PLA prototypes, testing each on an SFDI system at 851 nm and iterating. I helped coordinate fabrication timelines and test protocols and documented our methods and results. The result is an adjustable, controlled phantom that SFDI can measure at the target angles, a platform that could support future diagnosis and monitoring of skin fibrosis and improve the accuracy of optical imaging.



CAD model of the 3D-printed phantom



Printed phantom with angle-labeled tabs



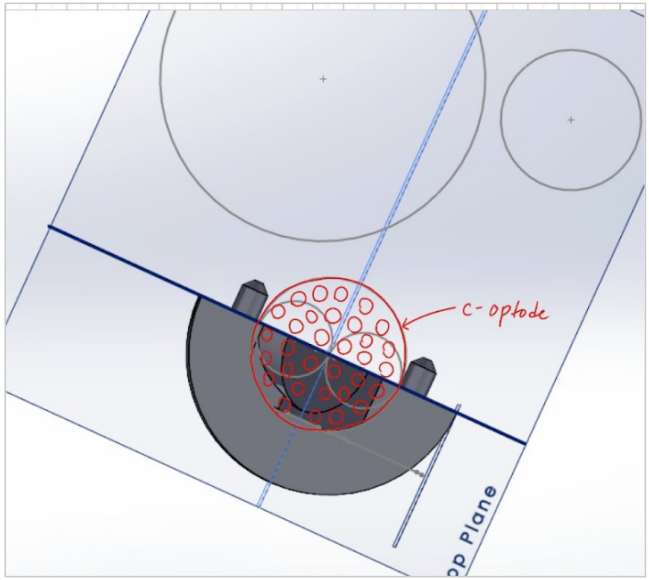
Slicer settings in Bambu Studio: infill direction, pattern, and density

Optical Probe (Optode) Casing for Blood Flow Measurement

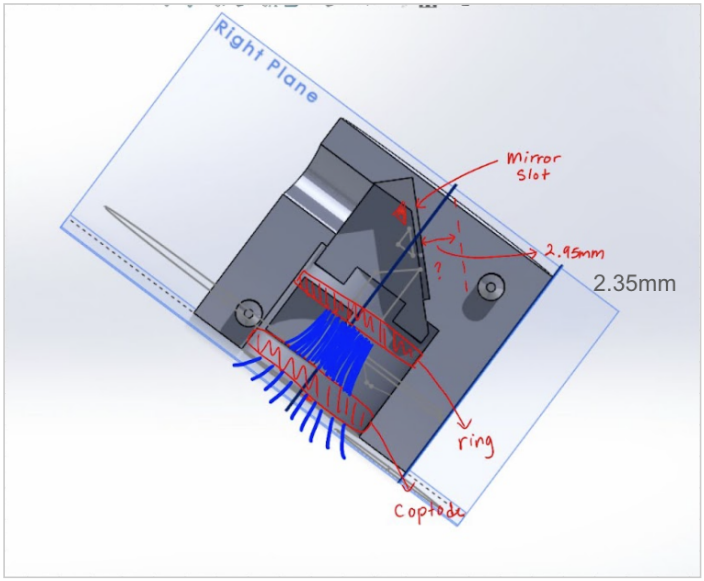
Bio Optical & Acoustic Spectroscopy Lab, Boston University · Oct 2024 – Oct 2025 | SolidWorks · Optics · Fabrication

Our spectroscopy lab measures blood flow with optical fibers, and measuring through hair, such as on a person who is not bald, requires small fibers bundled together. My task was to design a casing that holds that bundle of small fibers in place.

I designed and fabricated the casing in SolidWorks, including the fiber bundle layout, a mirror slot, and a ring, and iterated on imaging results. I also analyzed large imaging datasets in MATLAB and presented design trade-offs in weekly team meetings. The casing was designed to improve measurement accuracy by about 50%. We did not take the product further because it took too much time and cost, so we stopped after weighing that trade-off.



Fiber bundle layout sketched on the SolidWorks model (c-optode)



Annotated casing design: mirror slot, ring, and dimensions

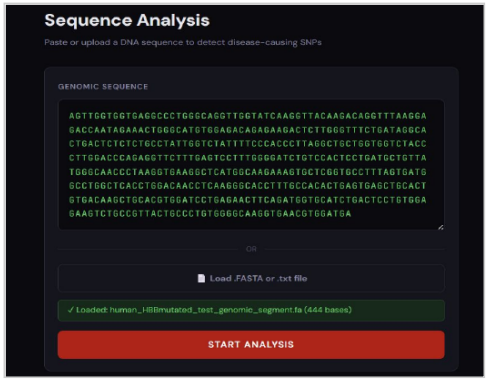
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Cleave SNP: SNP Detection and CRISPR Guide Design

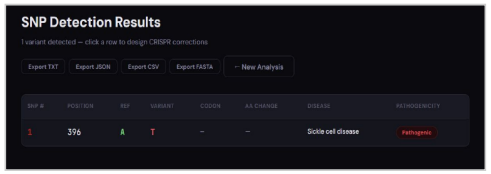
EC552 Computational Synthetic Biology, Boston University · Team of 4 · Apr 2026 | Python · BLAST · React · CRISPR

Most SNP tools stop at detecting a mutation, with no simple framework that carries a patient sequence from detection to genetic correction guidance. Our team of four set out to build a web platform, using sickle cell disease (the HBB gene) as the case study, that detects disease-associated SNPs and designs ranked CRISPR guide RNAs. Our specs were queries of at least 20 bp, 100% accuracy on validated test cases, and a workflow of 7 steps or fewer.

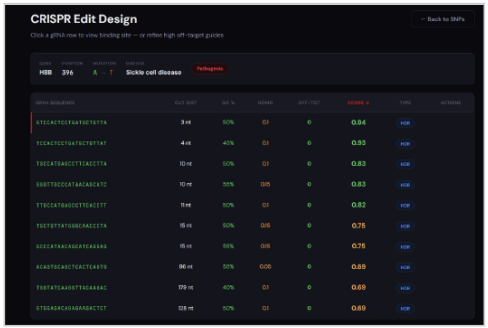
We built BLAST alignment with sequence safety checks, SNP calling matched to known disease variants, exports to TXT, JSON, CSV, and FASTA, a guide RNA step that scores PAM sites by SNP-to-cut distance, GC content, homopolymer penalty, and off-target activity, and a React web interface. When our own gRNA module could not be validated, we used Benchling for scoring and visualization. The app detected the sickle cell variant in a test HBB segment and flagged it as pathogenic, and the workflow takes 6 steps. We traced a position that was off by one to Python indexing starting at 0, and our next steps are validating the gRNA module against gold-standard tools and improving alignment filtering.



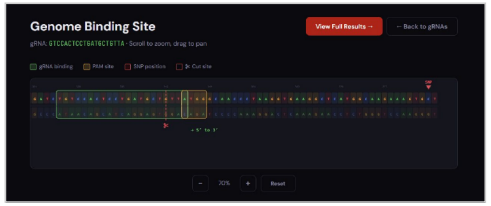
Sequence input for the HBB test segment



SNP detection result: sickle cell variant, pathogenic



CRISPR edit design: ranked guide RNAs



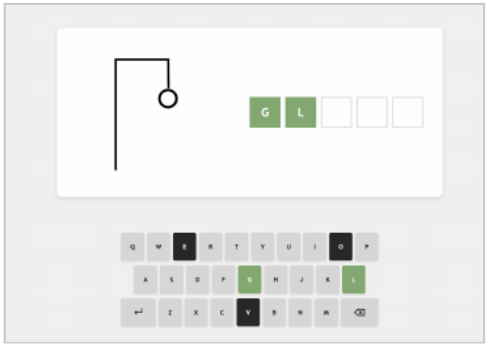
Genome binding site view for a selected gRNA

Hangle: Hangman Meets Wordle

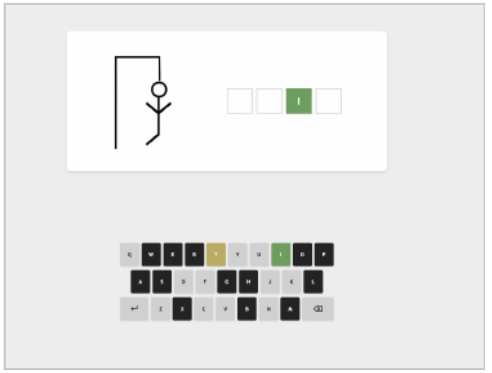
EC327 Introduction to Software Engineering, Boston University · Team of 6 · Fall 2024 | JavaScript · Git · Agile

Our software engineering course asked each team to build a web app, with a Hangman game that supports solo and two-player modes as the minimum and extra credit for added features. Our team of six set out to combine Hangman and Wordle into one polished game.

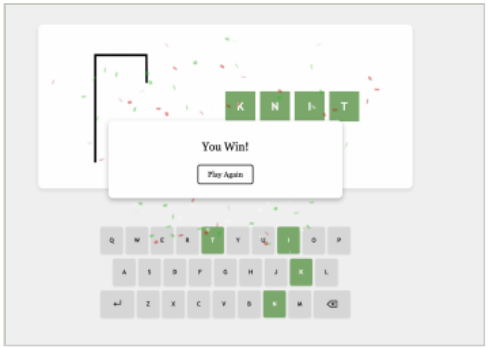
I implemented modular, object-oriented JavaScript for the game logic, animations, and user interaction, and the team used Git, agile practices, and code reviews. We delivered a working web app, presented with a live demo, with Easy, Medium, and Hard levels, a two-player mode, hints, extra lives, win and loss animations, dark mode, a challenge mode, and a sound toggle. It supports both letter guessing with a hanging man and Wordle-style word guessing.



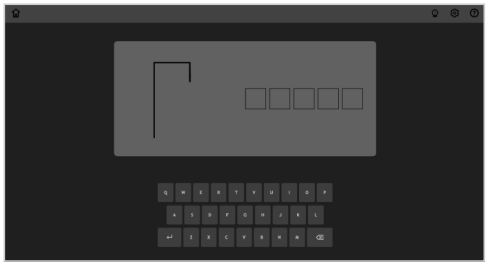
Wordle-style feedback: green for correct position



Hangman figure builds with each wrong guess



Win: confetti animation



Dark mode

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Additional Work

Photos and CAD renders available on request

Room Occupancy Monitor

Introduction to Engineering Design · Fall 2023

In a design course, I built a room occupancy monitoring system with alerts to meet defined requirements. I designed and prototyped it on Arduino and made the enclosure and electronics with laser cutting, soldering, and circuit wiring. The result was a working prototype tested against the requirements.

Effects of Caffeine on Heart Rate Recovery

Systems Physiology · Summer 2024

In a physiology course, I designed and ran a human-subject experiment on how caffeine affects heart rate recovery. We followed a defined protocol, collected data with LS4 software and plethysmography, and analyzed it with a two-way correlated ANOVA and paired t-tests, then reported the statistical significance of the results.